4.12 Wind Chill (Lab 4.1)

LAB ACTIVITY | 4.12.1: Wind Chill (Lab 4.1) | Full screen | 0 / 50

Determining how cold it feels outside is not the same as just knowing the current temperature. In this exercise we will use the following National Weather Service formula to determine the wind-chill temperature, which uses just the current temperature and windspeed.

$$t_{wc} = 35.74 + 0.6215t_a - 35.75v^{0.16} + 0.4275t_a v^{0.16}$$

Of course, other factors, such as relative humidity and amount of cloud cover, play important roles in determining how many layers you will want to put on before venturing outside, but we'll satisfy ourselves with temperature and wind speed.

First, download this template and rename it `wind_chill.py`:

- template.py

Then, implement the formula given above as part of this algorithm:

> **Prompt for and read** from the user the temperature in degrees Fahrenheit (called t_a in the formula) and the wind speed (called v in the formula) measured in miles per hour.
> **Compute** the wind-chill temperature (called t_{wc} in the formula).
> **Print** the result.

Here is an example execution.

```
Temperature: 0.0↵
Wind speed: 5.0↵
Wind chill: -10.509947791689996
```

At this point you should have a program that will compute the wind-chill temperature. For sample values, see the chart of wind chill values linked from Wikipedia's wind chill page (source NOAA). However, the formula given above cannot be used if the wind speed is below 2 mph or if the temperature is below -58 or above 41 degrees Fahrenheit.

Update your wind chill implementation to validate its input as follows.

> **Prompt for and read** from the user the temperature in degrees Fahrenheit (called t_a in the formula) and the wind speed (called v in the formula) measured in miles per hour.
> **If** the wind speed is less than 2 OR the temperature is less than -58 OR the temperature is greater than 41:
> **Print** a message saying the program will not work (using the message `Invalid input`).
> **Otherwise**
> **Compute** the wind-chill temperature (called t_{wc} in the formula)
> **Print** the result

We now have a program that will only do the computation when it is appropriate, but that still doesn't help us know how many layers to put on when we want to head outside. Let's fix this!

Update your program to print out both the wind-chill temperature and an indication of how many layers to wear. Use the following cutoffs:

- wc_temp < -40 : Stay home!
- -40 <= wc_temp < -10 : Four layers
- -10 <= wc_temp < 20 : Three layers
- 20 <= wc_temp < 40 : Two layers
- wc_temp >= 40 : One layer

Here are two example executions.

```
Temperature: 0.0↵
Wind speed: 5.0↵
Wind chill: -10.509947791689996
Four layers
```

```
Temperature: -100
Wind speed: 100
Invalid input
```

Open new tab | Dock

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Coding trail of your work What is this?

```
History of your effort will appear here once you begin working on this
zyLab.
```

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